A Mathematical-Consistency Verifier for a Phenomenological Curved-Spacetime Quantum-Channel Model

Executable Internal-Consistency Checks for the QSOT2 Line

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Abstract

We present `QSOT2`, a bounded *mathematical-consistency verifier* for a phenomenological model mapping classical curved-spacetime curvature norms (Frobenius norms of the Riemann and Ricci tensors) to single-qubit decoherence channels. The mapping is a conceptual ansatz with a free calibration parameter α; QSOT2 makes no first-principles or external-physical-validity claim. Its sole deliverable is to verify that the model holds together as quantum mechanics and as executable software: temporal-state axioms (linearity, CPTP completeness, trace preservation, conditionability, density validity) are checked to machine epsilon, channels are CPTP by construction, a flat-relative Kirkwood-Dirac comparison is reported as a bounded diagnostic, and a TTM-inspired trace-distance proxy is split into a detector self-test and the model trajectory. These are executable regression and internal-consistency tests for the implemented ansatz, not empirical evidence that the curvature-to-channel mapping is physically correct; the curvature norms themselves are non-derived toy proxies, so the product $\alpha\|\text{Riemann}\|_F$ is a pure number carrying no physical units. The reference run over six spacetime backgrounds emits a machine-readable result with a four-field claim boundary and an overall `DEGRADED_PASS` verdict across 35 deterministic checks — degraded for exactly one documented reason: the flat-baseline KD optimizer does not converge within its fixed step budget.

Reproducibility. One command runs the experiment and one runs the test suite (50 tests, ∼96% coverage). The retained numerical outputs reproduce the QSOT-Harness r1 math engine; the exact commands and parity targets are given in §8.

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Contents

1 Scope and Background

QSOT abbreviates *Quantum State Over Time.* `QSOT2` is a mathematical-consistency verifier for a phenomenological curved-spacetime quantum-channel model: it maps the Frobenius norm of a background's Riemann tensor to a single-qubit decoherence channel and checks that the resulting quantum-mechanical machinery is internally consistent, deterministic, and reproducible.

Lineage. The QSOT line passed through three forms. *QSOT v1* was an intentional high-formality slop artifact, formal-looking but effectively non-running. *QSOT-Harness* was an honest reconstruction that made the model run and added an epistemic-governance shell; it remains the published combined snapshot. `QSOT2` re-scopes that work to its mathematical core — the model and its verification harness — with the governance machinery deliberately lifted out so the verification is legible on its own terms [YI26].

Retained scope. QSOT2 verifies (i) internal quantum-mechanical consistency of the implemented maps, (ii) deterministic execution of an end-to-end pipeline, and (iii) reproducible numerical outputs. What it deliberately does *not* claim is centralized in §7.

Claim boundary. A passing QSOT2 check certifies model-internal mathematical consistency under stated assumptions *only*. The curvature-to-channel mapping is a phenomenological ansatz whose curvature inputs are non-derived toy values; the admissibility-gate and verdict labels are implementation-defined classifications, not physical observables. Accordingly the checks read as executable regression and consistency tests, not as empirical support for the ansatz. The full non-claim list is centralized in §7.

2 Phase 0: Temporal Axiom Contract Checks

Phase 0 verifies five temporal-state axioms of the implemented single-qubit channel maps. Each is a falsifiable numerical predicate evaluated to machine precision; a check passes when its deviation is below a fixed tolerance ($\texttt{atol} = \texttt{rtol} = 10^{-9}$). The axioms are standard consistency conditions for quantum channels and open-system state evolution [BP02].

Linearity. For a channel $\mathcal{E}$, states ρ_a, ρ_b, and scalars p, q, the quantity $\|\mathcal{E}(p\rho_a + q\rho_b) - p\,\mathcal{E}(\rho_a) - q\,\mathcal{E}(\rho_b)\|$ must vanish.

CPTP completeness. For Kraus operators $\{K_i\}$, $\|\sum_i K_i^\dagger K_i - I\|_F$ must vanish.

Trace preservation. For each test state, $|\operatorname{Tr}\mathcal{E}(\rho) - \operatorname{Tr}\rho|$ must vanish.

Conditionability. One- and two-step trajectory replays must match the composed channel prediction: $\rho_{i+1} \approx \mathcal{E}_i(\rho_i)$ and $\rho_{i+2} \approx \mathcal{E}_{i+1}(\mathcal{E}_i(\rho_i))$.

Density validity. Each evolved ρ must be Hermitian, unit-trace, and positive semidefinite.

In the reference run the realized deviations sit at machine epsilon: linearity $\sim 1.1 \times 10^{-16}$, CPTP completeness $\sim 1.6 \times 10^{-16}$, and trace preservation $\sim 4.4 \times 10^{-16}$; the composed-channel, sequential-replay, and two-step conditionability deviations are exactly 0. These are reported as machine-precision evidence, not as prose assurances. Because the channels are CPTP by construction, these axiom checks function as executable regression and consistency tests: they guard the implementation against drift and confirm the maps behave as specified, but they are *not* independent evidence that the underlying ansatz is physically correct. That distinction is the point — Phase 0 makes QSOT2 a verifier with falsifiable numerical predicates rather than a narrative artifact.

3 Phases 1–3: Baselines, Curvature Noise, and Boosts

Phases 1–3 exercise the retained model pipeline outside the KD and memory-proxy diagnostics.

Flat baselines (Phase 1). A qubit evolved through a flat Minkowski background, at rest and boosted, preserves purity ($P = 1.0$) and zero von Neumann entropy and clears the causal-structure gate. This fixes the no-noise limit of the implemented map.

Curvature noise (Phase 2). Across the curved backgrounds the channel map $p = 1 - \exp(-\alpha\|\mathrm{Riemann}\|_F)$ drives purity decay and entropy growth. Here $\|\mathrm{Riemann}\|_F$ is the Frobenius norm of a *non-derived* curvature proxy — hand-set toy constants for the Ricci-flat cases and a copy of the Ricci matrix for the conformally curved cases (each background records its `riemann_source` and a boolean `riemann_is_toy` flag in the result artifact). The norm is therefore a pure number, so $\alpha\|\mathrm{Riemann}\|_F$ is dimensionless by construction and α is a dimensionless calibration multiplier, not a physical curvature scale. In the broader literature, gravity-related time dilation and curvature have long been discussed as potential decoherence mechanisms [Pik+15]; here that connection is retained only as phenomenological motivation, not as a derivation. In the reference run ($\alpha = 0.1$): Schwarzschild $P = 0.99943$, de Sitter $P = 0.63607$, AdS5 $P = 0.67764$, and Eguchi–Hanson $P = 0.99887$, with von Neumann entropies of 0.0026, 0.5501, 0.5031, and 0.0048 respectively. The design separates the two curvature roles: the *Riemann* norm drives decoherence — so Ricci-flat vacua such as Schwarzschild and Eguchi–Hanson still decohere through non-zero tidal curvature — while the *Ricci*-based residual feeds the admissibility gate. The retained gate pattern is `PASS` for flat/Schwarzschild/AdS5 and `FAIL` for de Sitter (non-zero beta residual), Eguchi–Hanson (gravitational-anomaly proxy), and Gödel (closed timelike curve). These are implementation-defined classifications, not physical observables; each verify block serializes its decision as a machine-readable `policy_reason_code` field (e.g. `nonzero_beta_residual`, `ctc_failure`, `anomaly_failure`).

Relativistic boosts (Phase 3). A boost $\beta > 0$ increases the effective noise parameter through the combination of gravitational and kinematical time dilation, in the intended direction. This section explains how the implemented model behaves, not why nature must behave that way.

4 Phase 4: Flat-Relative KD Signal

Phase 4 reports a single bounded diagnostic and keeps its interpretation narrow.

1. The implementation searches single-qubit measurement-basis angles to minimize the real part of a Kirkwood-Dirac quasiprobability, $\min \mathrm{Re}\,\mathrm{Tr}(P_b P_a \rho)$, following the standard Kirkwood–Dirac quasiprobability construction [Kir33; Dir45].
2. The flat and de Sitter optima are reported separately: in the reference run $Q_{\mathrm{KD}}^{\mathrm{flat}} \approx -0.1234$ and $Q_{\mathrm{KD}}^{\mathrm{de\ Sitter}} \approx -4.7 \times 10^{-4}$.
3. The reported signal is the flat-relative difference $\Delta_{\mathrm{KD}} = Q_{\mathrm{KD}}^{\mathrm{de\ Sitter}} - Q_{\mathrm{KD}}^{\mathrm{flat}} \approx +0.1230$.
4. Convergence is reported independently of the value. Convergence here means the early-stop criterion fired: the objective failed to improve by more than 10^{-6} for 20 consecutive steps. On the flat baseline the objective is still improving at the final step (its best value coincides with the last step), so within the fixed 50-step budget the criterion never fires and convergence is recorded `false`; the de Sitter search, by contrast, early-stops at step 42. Non-convergence here therefore means *not yet plateaued*, not *diverged*: the reported value is the best observed objective and is well defined regardless. The KD check is surfaced as `DEGRADED_PASS` rather than hidden.

Two rules bound the reading. First, negative KD in an optimized basis is a generic property of single-qubit states; it is *not* a curvature or contextuality proof. Only the flat-relative delta is reported, and even that is a model-relative quantity. Second, the reported value tracks the *best* observed objective (best and last values are recorded separately), not merely the final optimizer step. This section therefore reads as a bounded diagnostic report, not a contextuality theorem. Phase 4 is a *diagnostic-tier* check: its convergence flag is the sole input to the single degraded label and is reported separately from the Phase 0 consistency axioms, which it does not affect (see the tiered reading in §6).

5 Phase 5: TTM-Inspired Memory Proxy

Phase 5 runs a TTM-inspired trace-distance proxy: at each step it accumulates the trace distance between the actual trajectory and the one-step Markovian channel prediction. It is a bounded backflow diagnostic, not a transfer-tensor reconstruction. The naming is deliberately comparative: the full Transfer Tensor Method reconstructs non-Markovian memory through transfer tensors inferred from process data [CC14], whereas QSOT2 uses only a trace-distance proxy in that spirit.

Two trajectories are kept distinct:

- *Synthetic self-test.* An artificial coherence backflow is injected to confirm the detector registers $\mathrm{nm} > 0$ (reference value $\mathrm{nm} \approx 1.4 \times 10^{-4}$ at $\alpha = 0.1$). A pass validates the detector only.
- *Model-generated trajectory.* Produced by applying the configured channel consistently with no injected revival, this yields $\mathrm{nm} = 0$: the phenomenological map is Markovian by construction.

The detection threshold and the Ricci norm at which it was evaluated are surfaced with the result, so the (here non-engaged) curvature scaling is explicit rather than implied. A zero model-trajectory result is a property of the implemented map, not a deep physical discovery.

6 Consolidated Results

Table 1 consolidates the reference run ($\alpha = 0.1$, six backgrounds). The pipeline runs Phase 0 plus Phases 1–5 (35 deterministic checks) and returns an overall `DEGRADED_PASS`: 34 checks pass and one is degraded (the flat-baseline KD non-convergence). There are no failures and no governance/audit phases. Read by tier, all five Phase 0 consistency axioms and every Phase 1–3 model check pass; the sole degraded entry is the diagnostic-tier KD convergence flag (§4), so all retained mathematical-consistency checks pass; the degraded label comes solely from optimizer hygiene, not an axiom violation.

Quantity	Value	Note
Overall verdict	`DEGRADED_PASS`	34 pass / 0 fail / 0 skip / 1 degraded
Checks (total)	35	Phase 0 + Phases 1–5
$P_{\mathrm{Schw}}/P_{\mathrm{dS}}/P_{\mathrm{AdS5}}/P_{\mathrm{EH}}$	0.99943/0.63607/0.67764/0.99887	purity (Schw / dS / AdS5 / EH)
$S_{\mathrm{dS}}/S_{\mathrm{AdS5}}$	0.5501/0.5031	von Neumann entropy
Δ_{KD}	+0.1230	diagnostic only; flat-relative, non-converged
nm (self-test / model)	1.41×10^{-4} / 0	detector vs. model
axiom deviations	$\leq 4.4 \times 10^{-16}$	machine epsilon

Table 1: Consolidated QSOT2 reference-run outputs.

Sensitivity. A three-point α sweep (committed as `reports/sweep_alpha.json`, regenerated by `scripts/alpha_sweep.py`) tests direction and ceiling, *not robustness*: the apparent plateau is a structural artifact of the flat baseline being curvature-free ($\|\mathrm{Riemann}\|_F = 0$), so its channel is the identity and its KD value $Q_{\mathrm{KD}}^{\mathrm{flat}} = -0.1234$ is *independent of* α. Only de Sitter responds to α: as it decoheres toward the maximally mixed state its KD optimum approaches 0, so Δ_{KD} rises toward the fixed ceiling $|Q_{\mathrm{KD}}^{\mathrm{flat}}| \approx 0.1234$ and is all but saturated by $\alpha = 1.0$. The measured values are $\Delta_{\mathrm{KD}} = +0.0135,\ +0.1230,\ +0.1235$ at $\alpha \in \{0.01, 0.1, 1.0\}$, with the model-trajectory memory result staying 0 throughout; absolute magnitudes remain calibration-dependent.

Parity. Every retained mathematical quantity above reproduces the QSOT-Harness r1 math engine numerically [YI26]. QSOT2 differs only by design: the governance/audit surfaces are absent and the result schema is leaner.

7 Limits and Explicit Non-Claims

This section centralizes the non-claims; the rest of the paper does not repeat them.

- No first-principles derivation of the curvature-to-channel map; it is a phenomenological ansatz.
- No claim that the curvature inputs are physically derived: the Riemann/Ricci norms are non-derived toy proxies (hand-set constants or Ricci copies, flagged per background by `riemann_is_toy`), so $\alpha\|\mathrm{Riemann}\|_F$ is dimensionless and carries no physical units.
- No external physical validation of the model.
- No claim that the free calibration parameter α is physically derived; it is a demonstrative mid-range setting, and quantitative magnitudes are calibration-dependent.
- No claim that the TTM-inspired proxy is a full transfer-tensor reconstruction.
- No governance-heavy audit interpretation in this paper; gate and verdict labels are implementation-defined classifications (serialized as `policy_reason_code`), and the epistemic-governance machinery is out of scope for QSOT2.

A passing check therefore means model-internal mathematical consistency under the stated assumptions — nothing more. Centralizing these non-claims here is a deliberate stylistic correction: in the predecessor line they had spread until they dominated every section.

8 Reproducibility

Run the experiment.

```
python run_experiment.py --config configs/experiment.yaml --out reports
```

Run the tests.

```
python -m pytest
```

The suite has 50 tests behind a 90% coverage gate; the current suite measures ∼96%.

Artifact	Role in paper	Schema / version	Traceability anchor
`reports/result.json`	Reference-run verdict, 35 checks, claim boundary, and retained numerical outputs	`qsot2.math_consistency.experiment_result.v1`	SHA-256 `efca2228f177634f8f0f83b93eb5a7bc9246afe55cf707e780d2791b8a685bfc`; release DOI `10.5281/zenodo.20763331`
`reports/sweep_alpha.json`	Three-point α sensitivity sweep behind §6	`qsot2.math_consistency.alpha_sweep.v1`	SHA-256 `8e547946e4b6d874ec58e23584977b07151571742be37d5e51da075d66bce7b6`
`configs/experiment.yaml`	Fixed reference-run configuration used by `run_experiment.py`	`version:0.1.2`	SHA-256 `0b5cd5f148d8d72b867fa343f0c364a77aaefe2465431ef5f590e0f15c91813e`
`pyproject.toml`	Environment and dependency contract for the published package line	`project.version=0.1.2`	SHA-256 `1d0bcd391d10c5194fee00d05c1f15d03797e044653bd6355a69f16ebb7f05bc`
`tests/`	Executable verification suite covering the retained math line	50 tests; coverage gate $\geq 90\%$	Repository test surface for the v0.1.2 line; current suite measures $\sim$96%

Table 2: Artifact traceability for the QSOT2 paper line.

Artifacts and schema. The run writes `reports/result.json` and `reports/report.md`. The result carries `schema_id = qsot2.math_consistency.experiment_result.v1` and a four-field claim boundary; it contains no evidence-class, audit-context, or calibration blocks. Some legacy field names remain in the serialized verification artifact for backward compatibility with the QSOT-Harness result surface; in QSOT2 they function only as implementation-defined classification metadata and are not interpreted as governance or audit evidence.

Artifact traceability. Table 2 ties each manuscript-facing claim surface to a concrete repository artifact. The stable DOI-bearing public release anchor for this paper line remains QSOT2 v0.1.1, DOI 10.5281/zenodo.20763331; the current repository line is v0.1.2. The hashes below are canonical-LF SHA-256 values of the repository files named in the first column.

Alpha sweep. The sensitivity sweep of §6 is reproduced by

```
python scripts/alpha_sweep.py
```

which writes `reports/sweep_alpha.json` (`schema_id = qsot2.math_consistency.alpha_sweep.v1`) without overwriting the frozen reference `reports/result.json`.

Parity targets. Against the QSOT-Harness r1 math engine the retained quantities reproduce: $Q_{\mathrm{KD}}^{\mathrm{flat}} \approx -0.1234$, $Q_{\mathrm{KD}}^{\mathrm{de\ Sitter}} \approx -4.7 \times 10^{-4}$, $\Delta_{\mathrm{KD}} \approx +0.1230$, self-test nm $\approx 1.41 \times 10^{-4}$, model nm $= 0$, purities 0.99943/0.63607/0.67764/0.99887, and axiom deviations at machine epsilon. Re-runs are content-stable except for the `generated_at` timestamp.

References

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